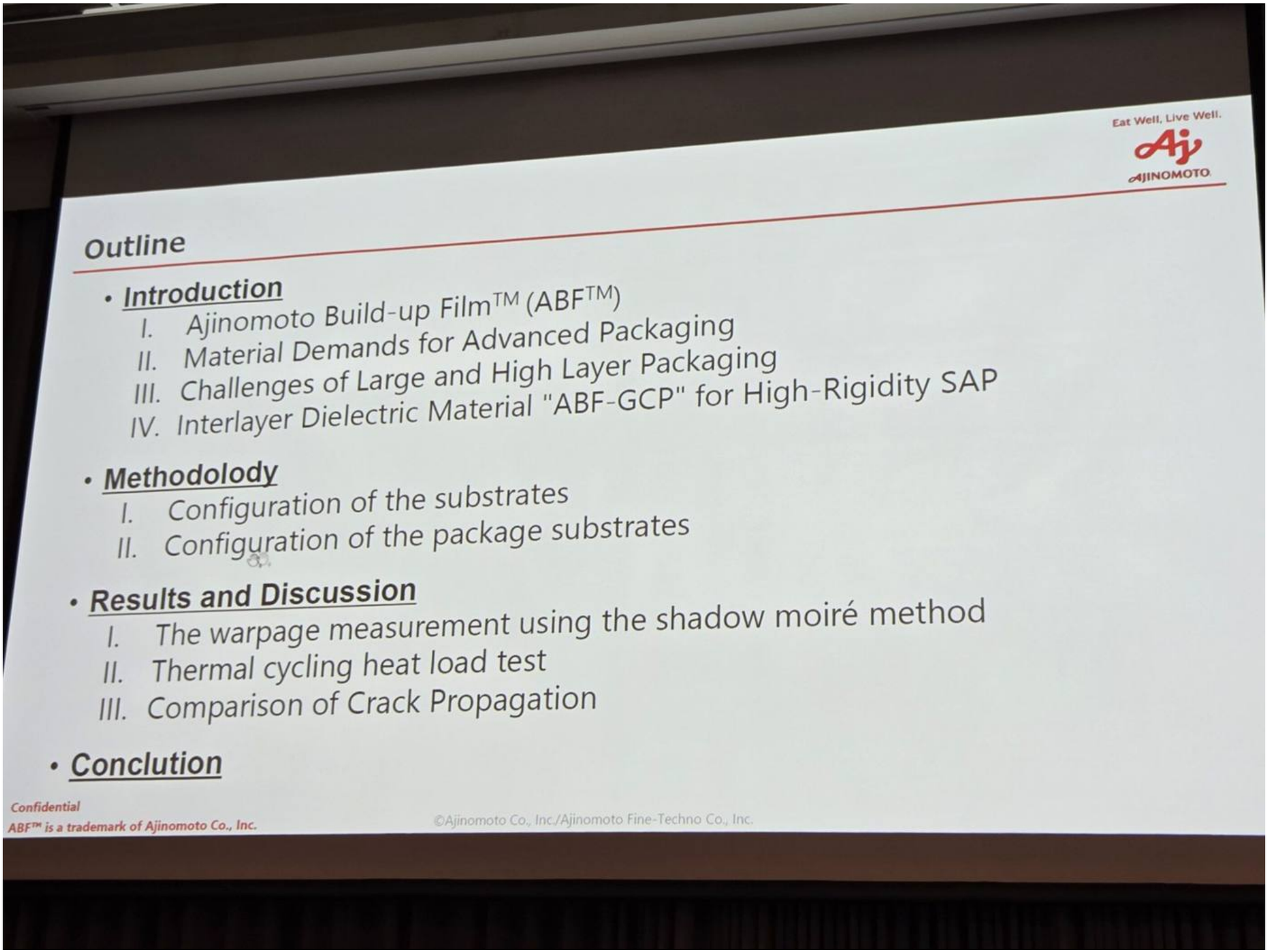
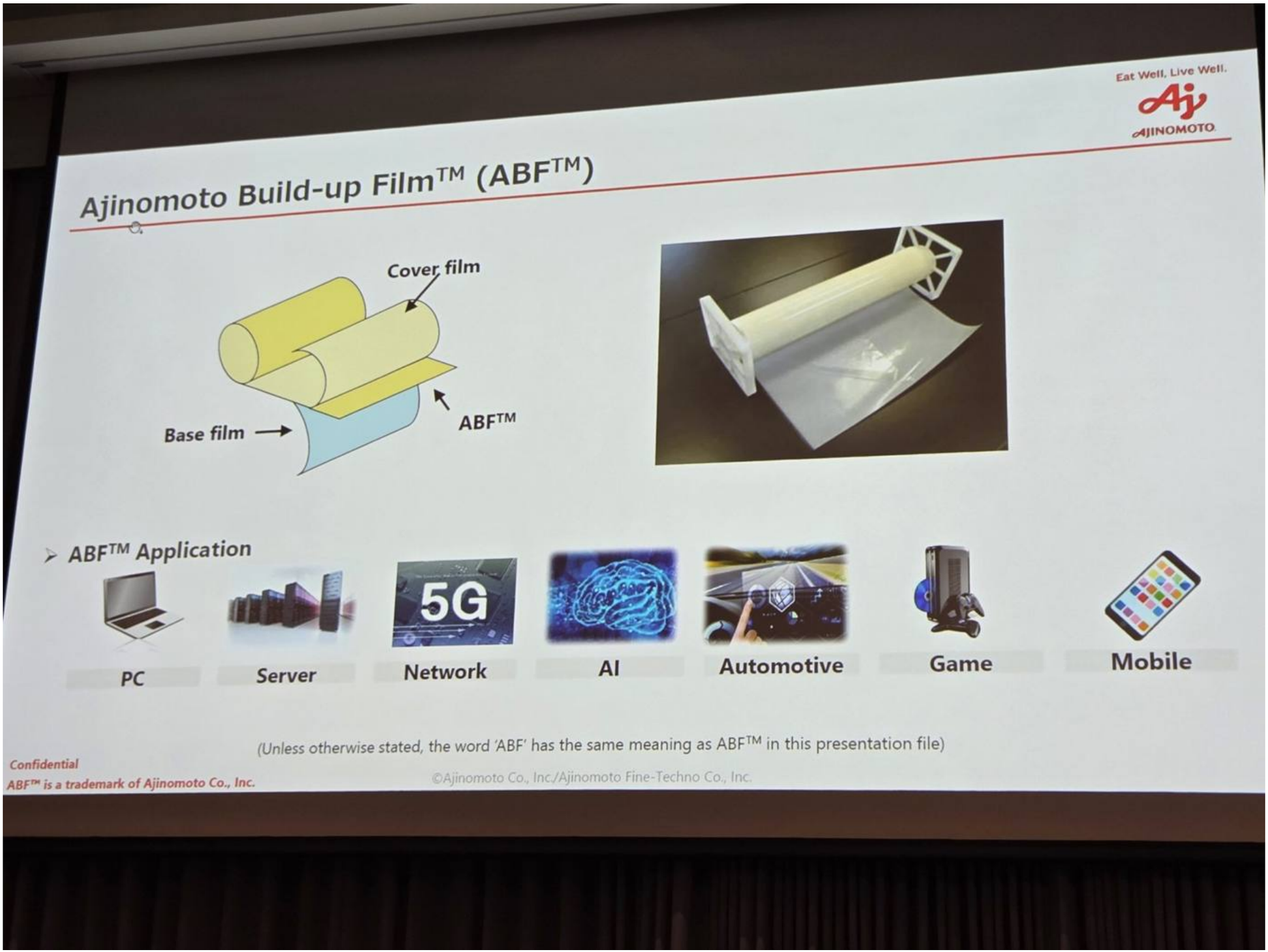
T21 유리섬유 보강 ABF의 휨과 크랙 제어

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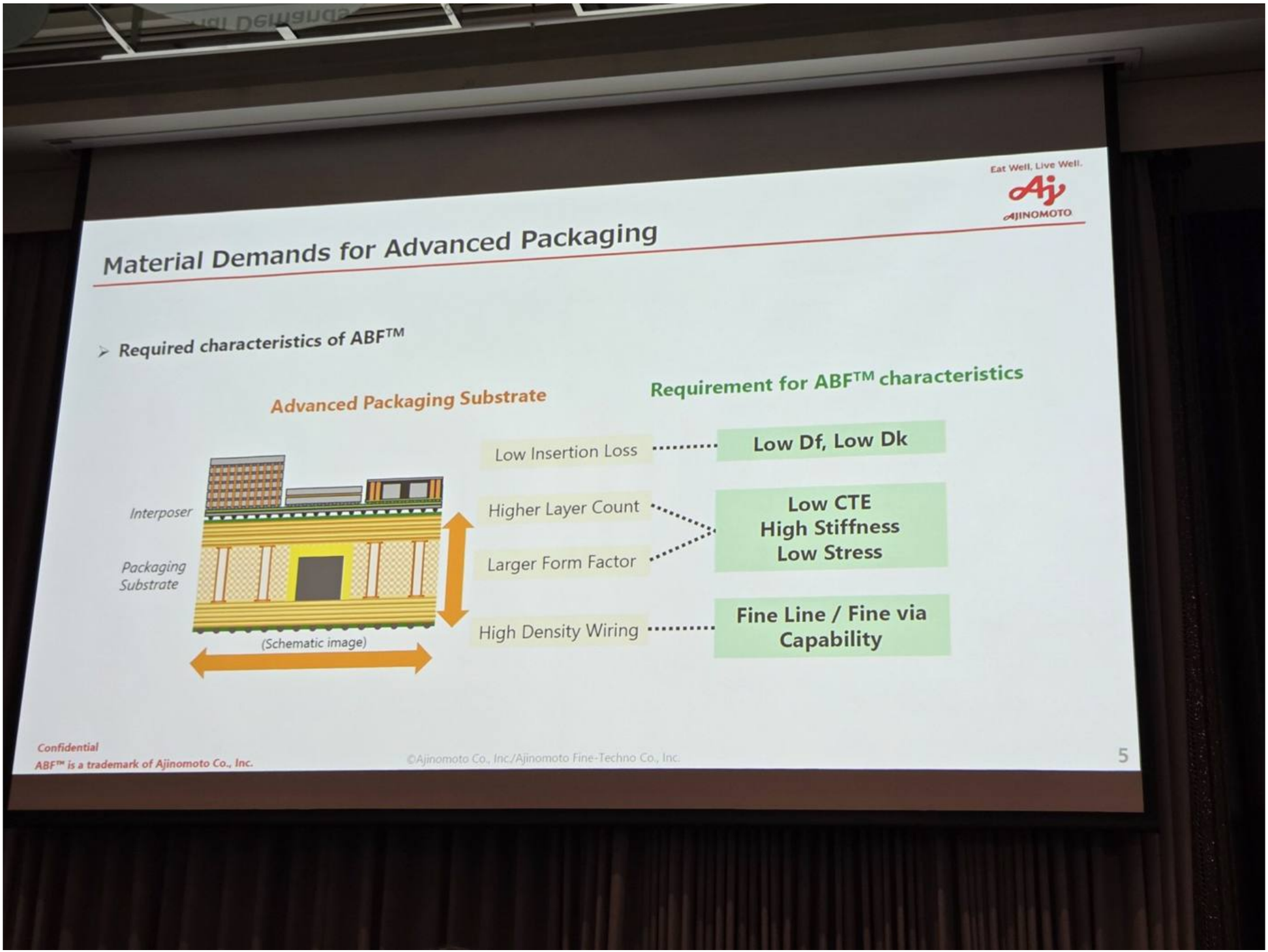
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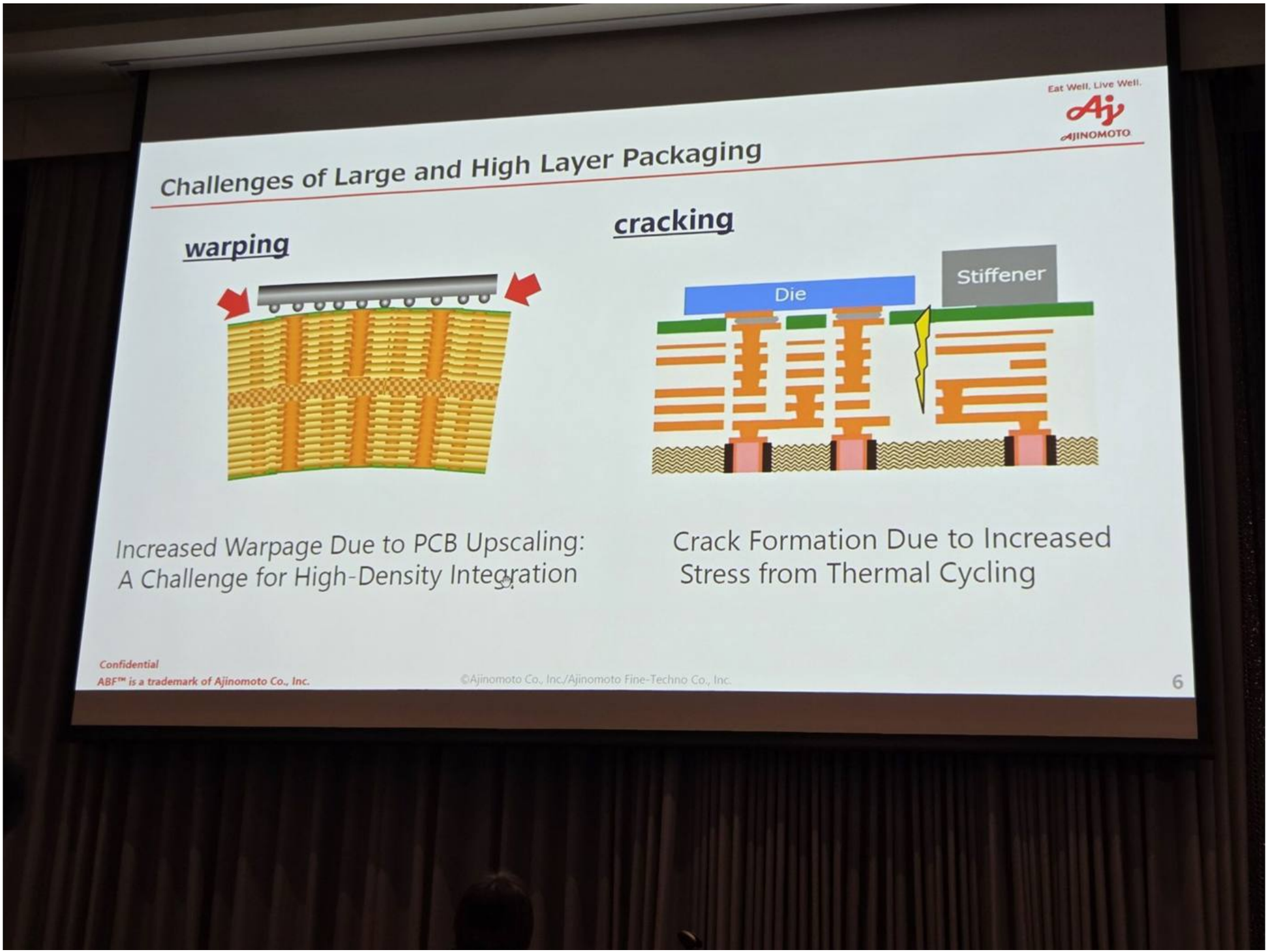
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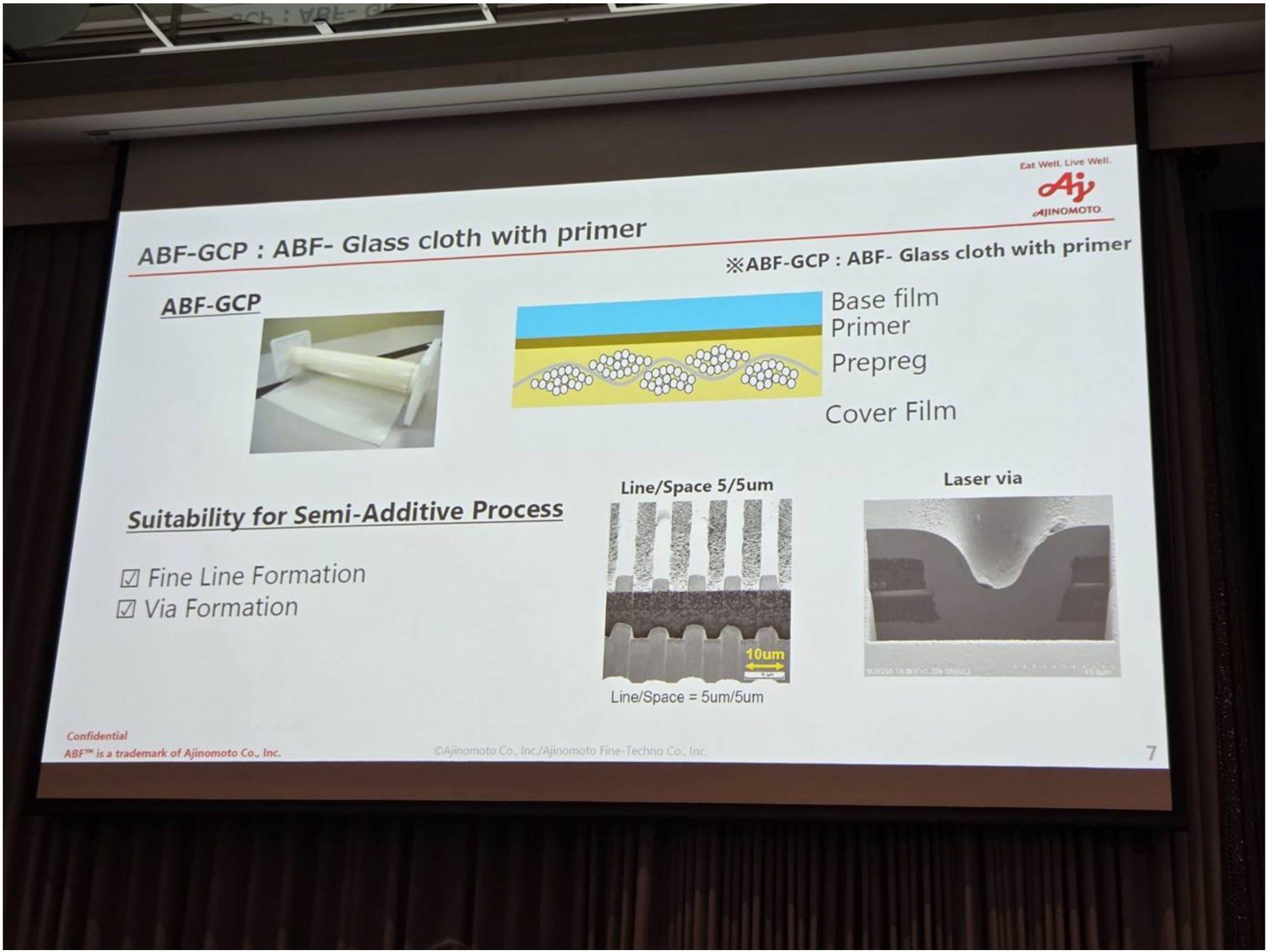
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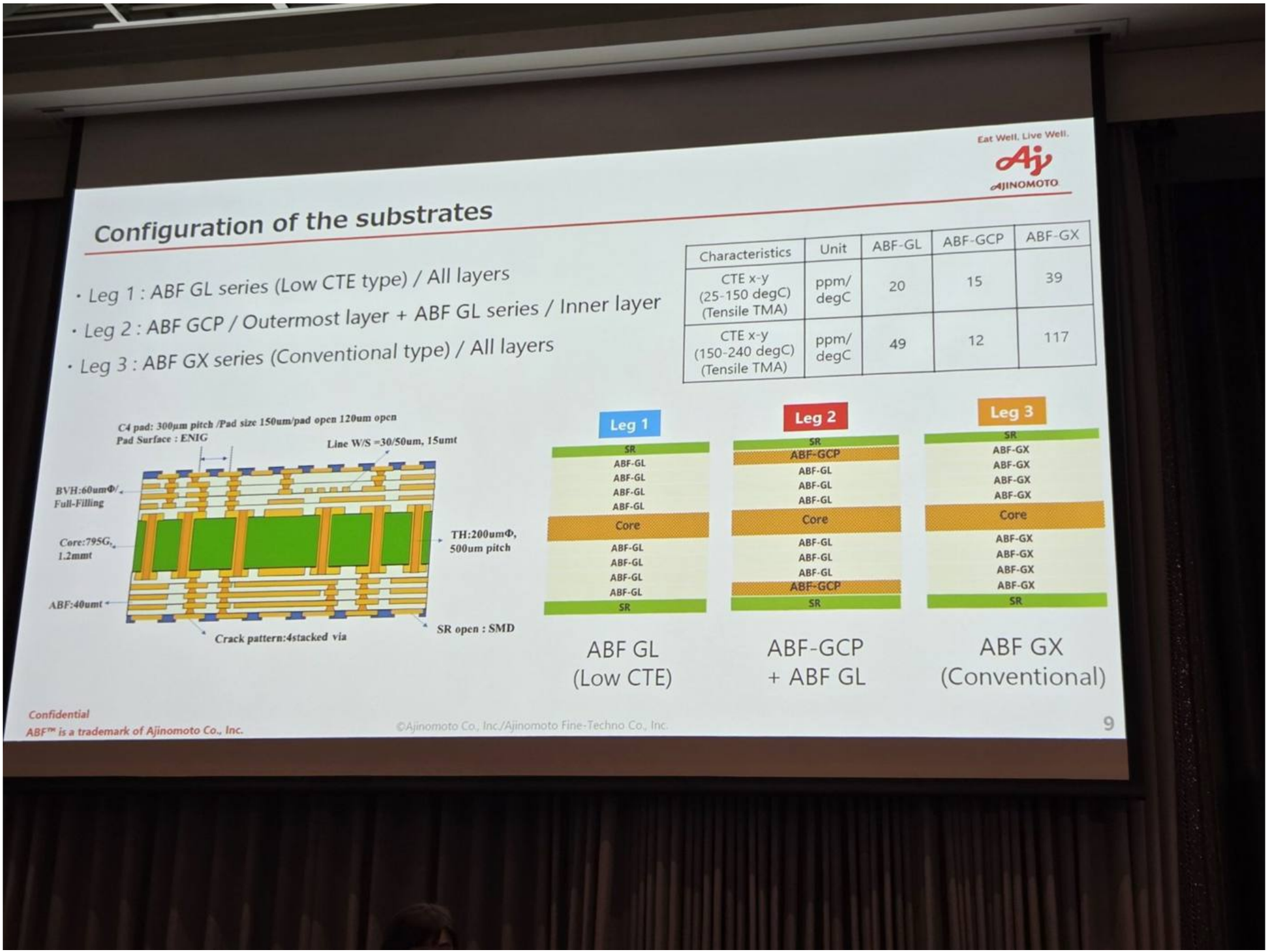
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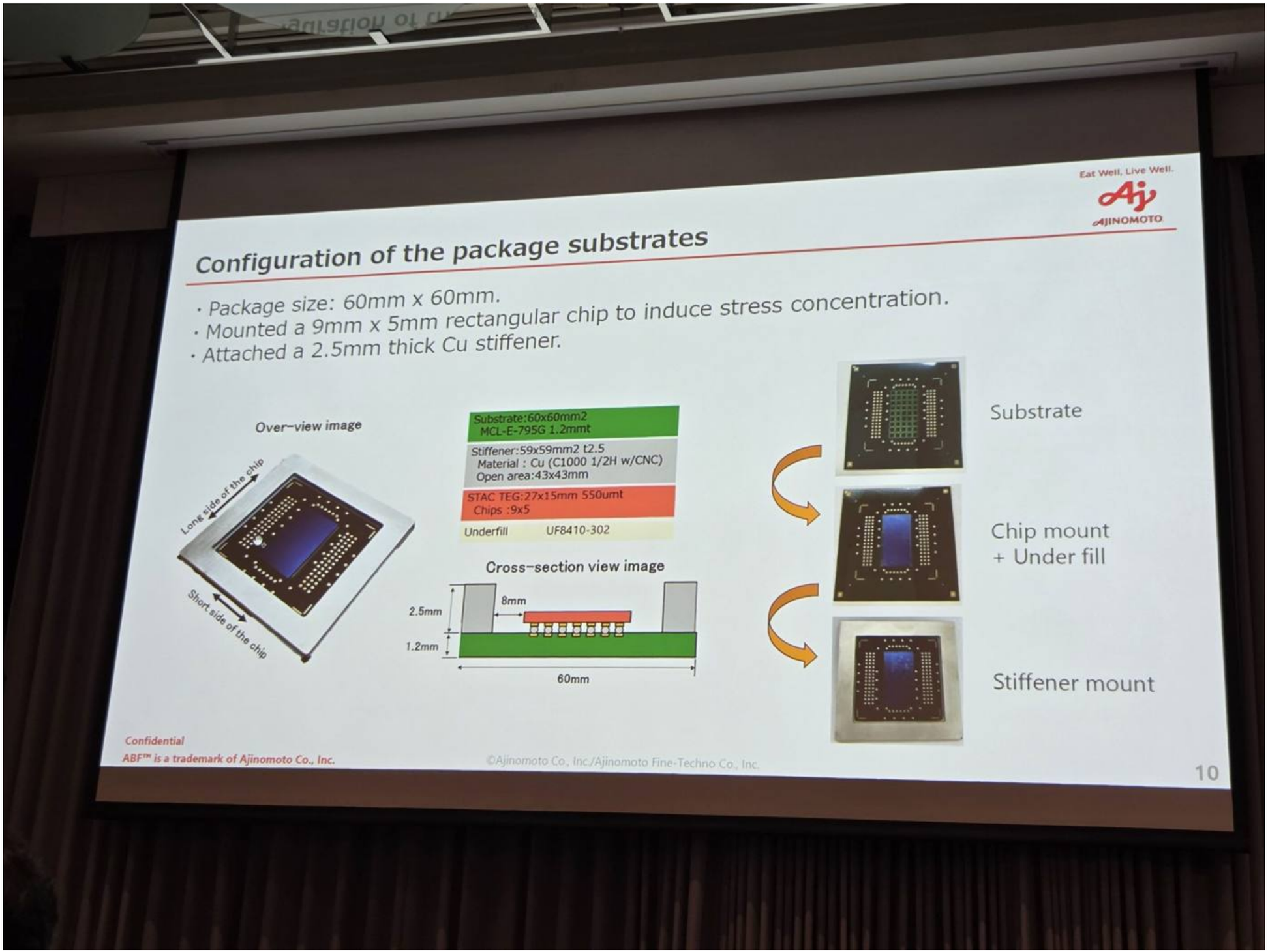
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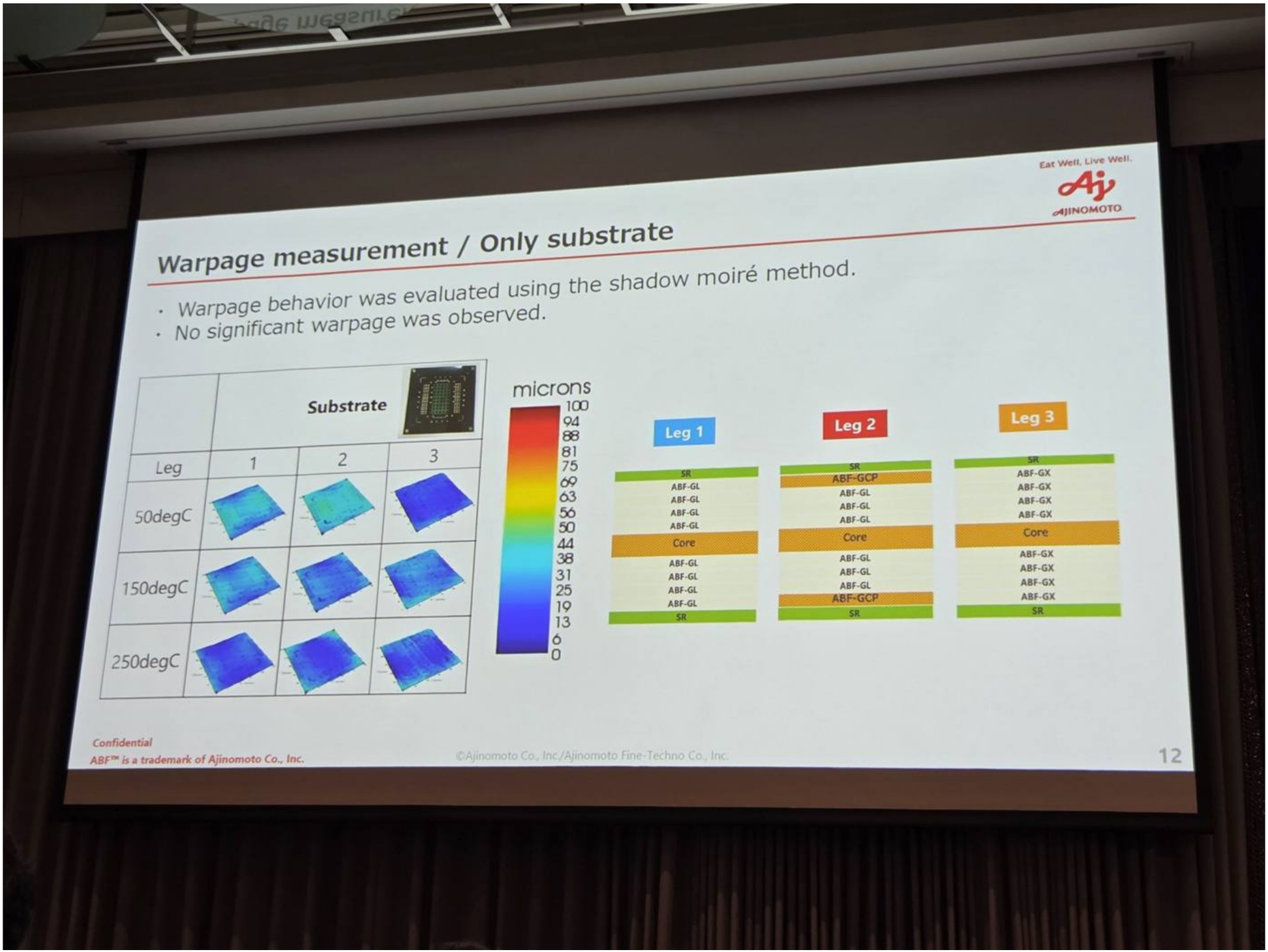
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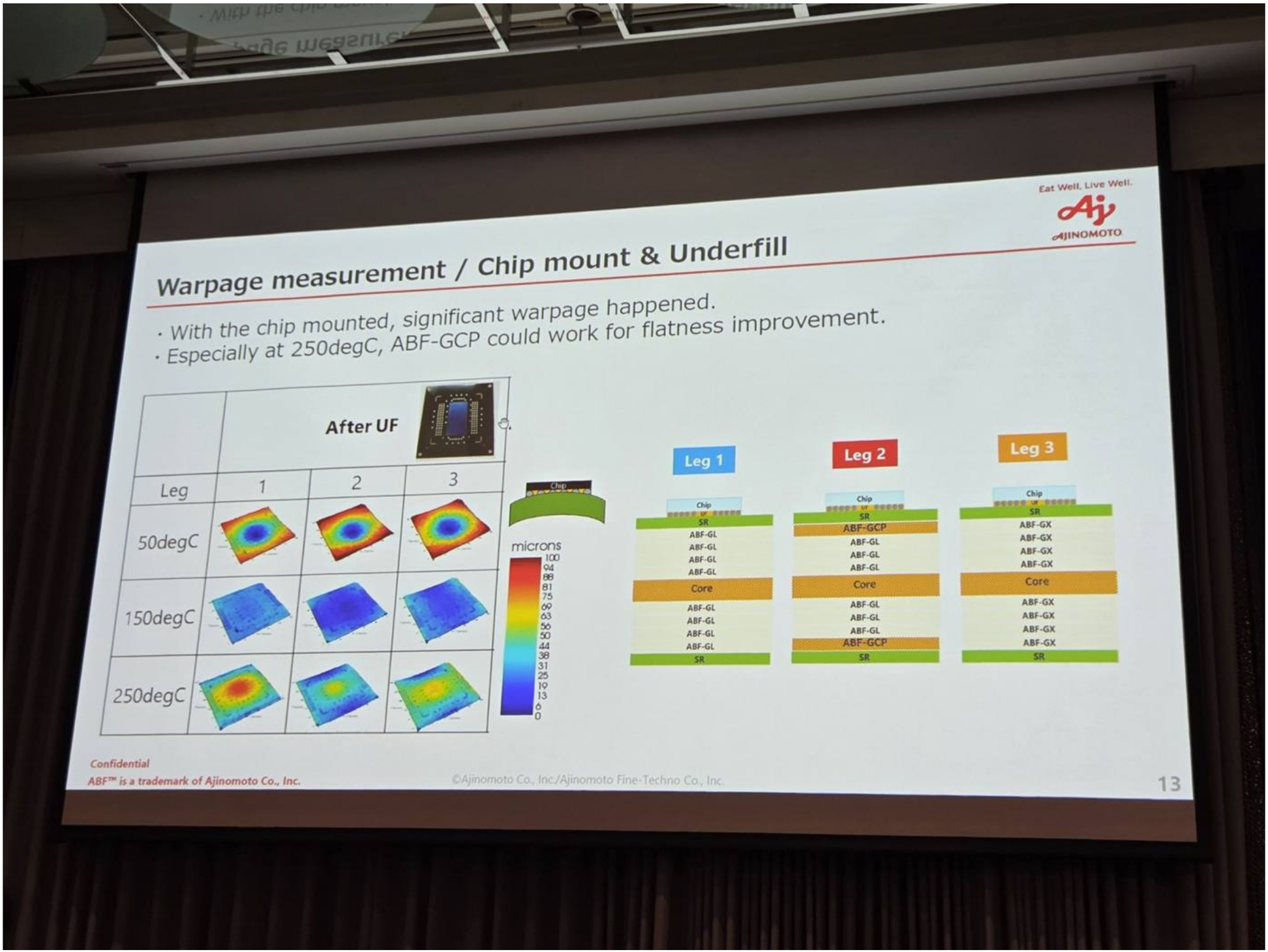
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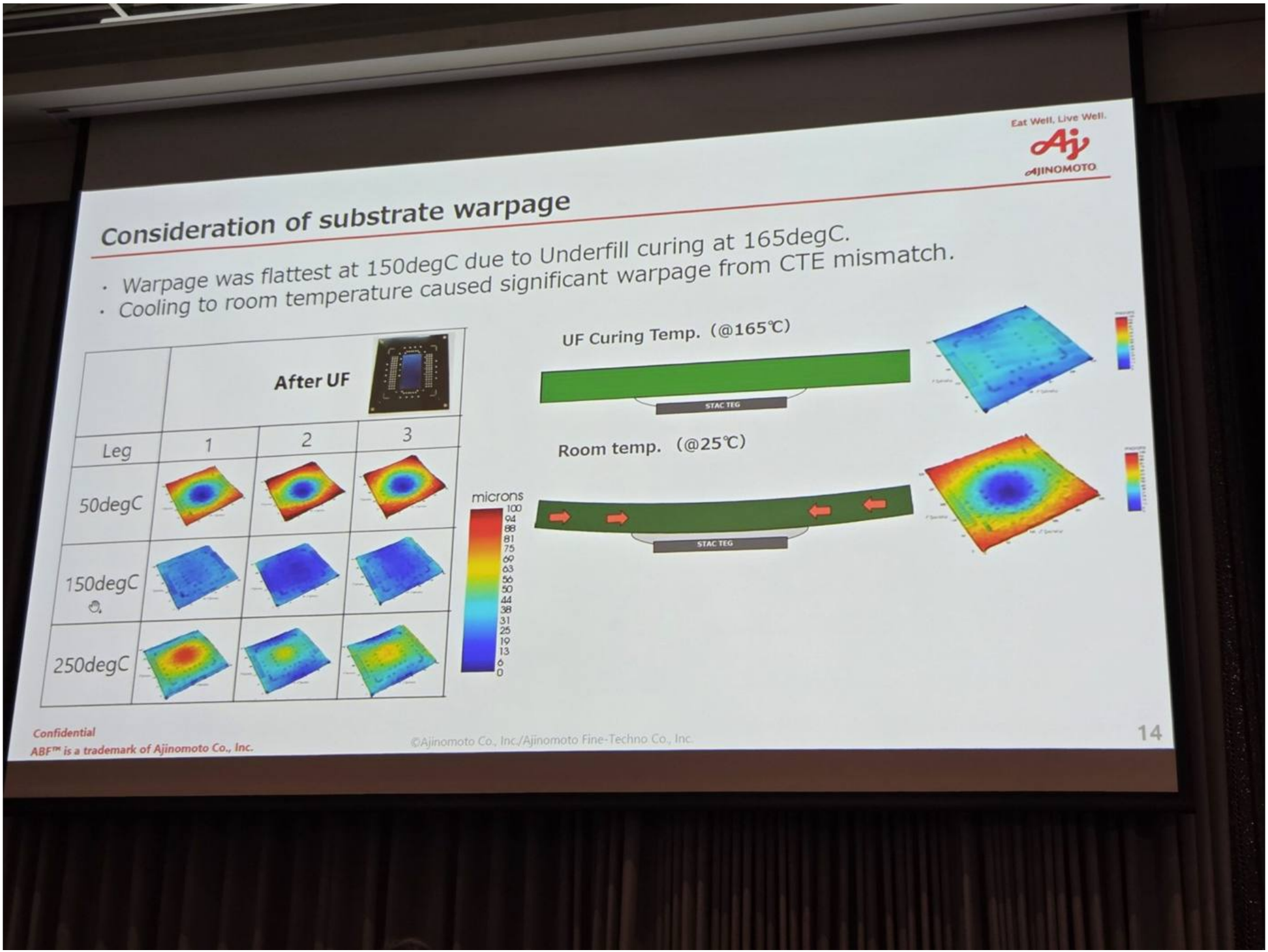
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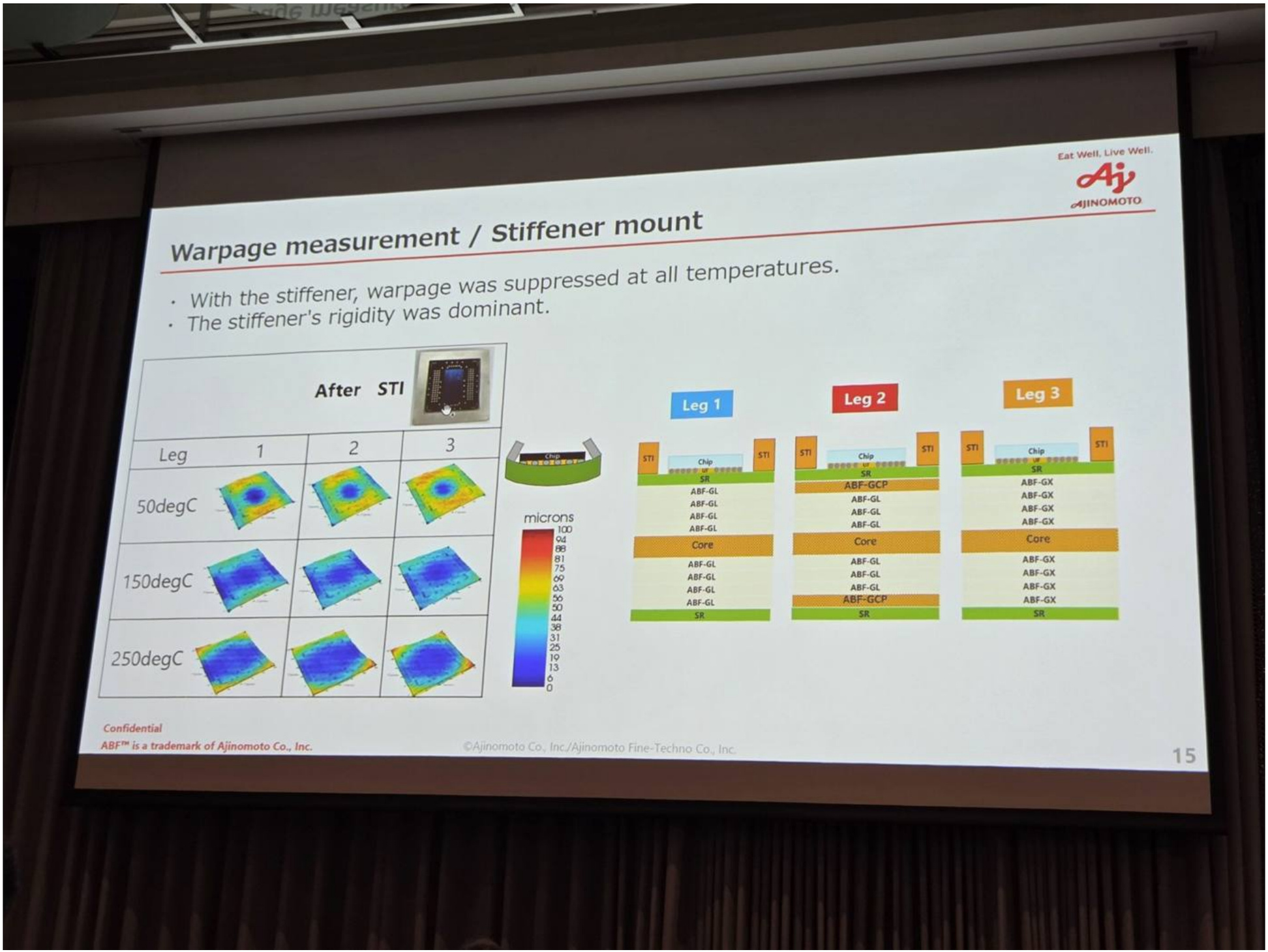
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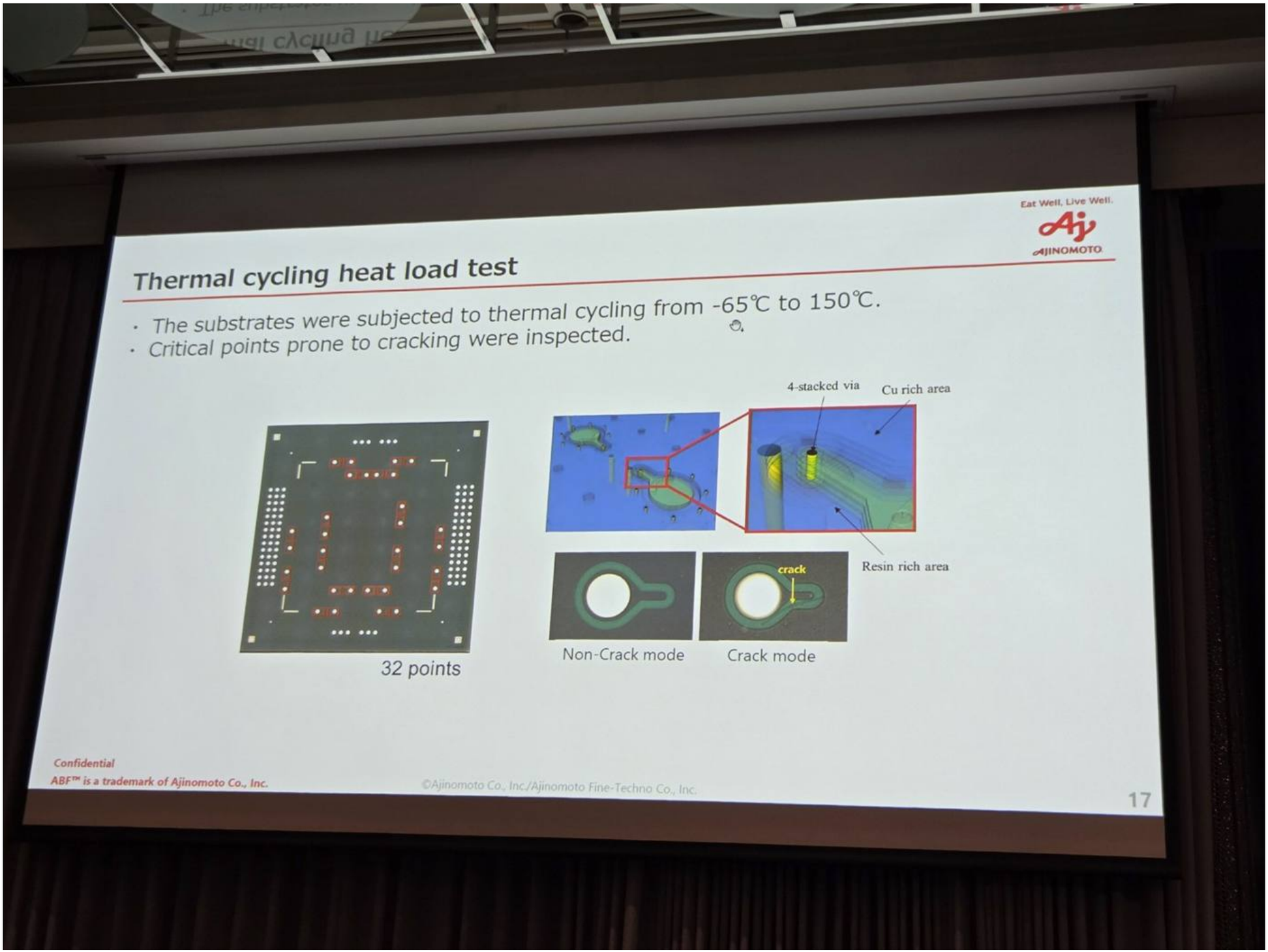
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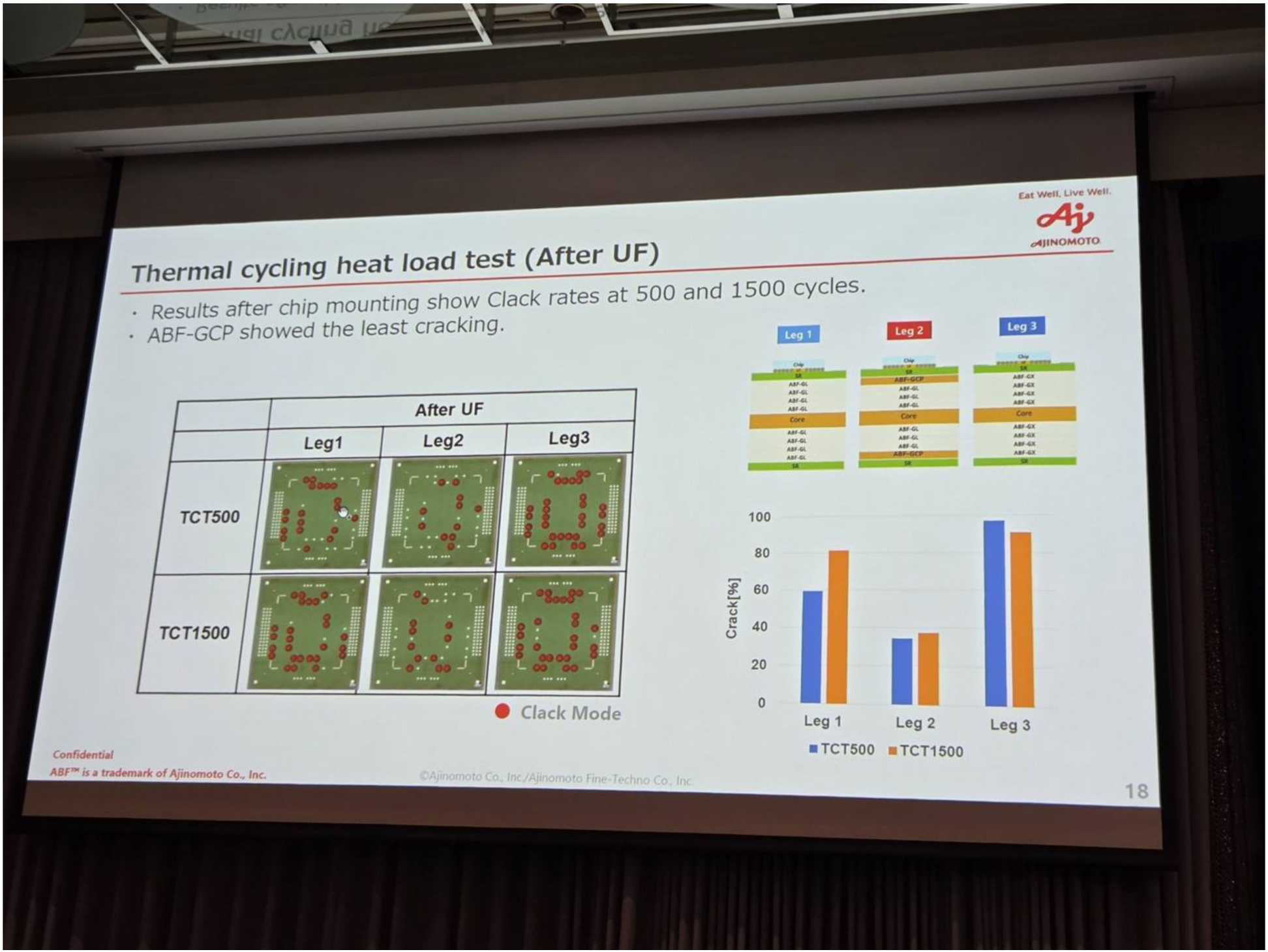
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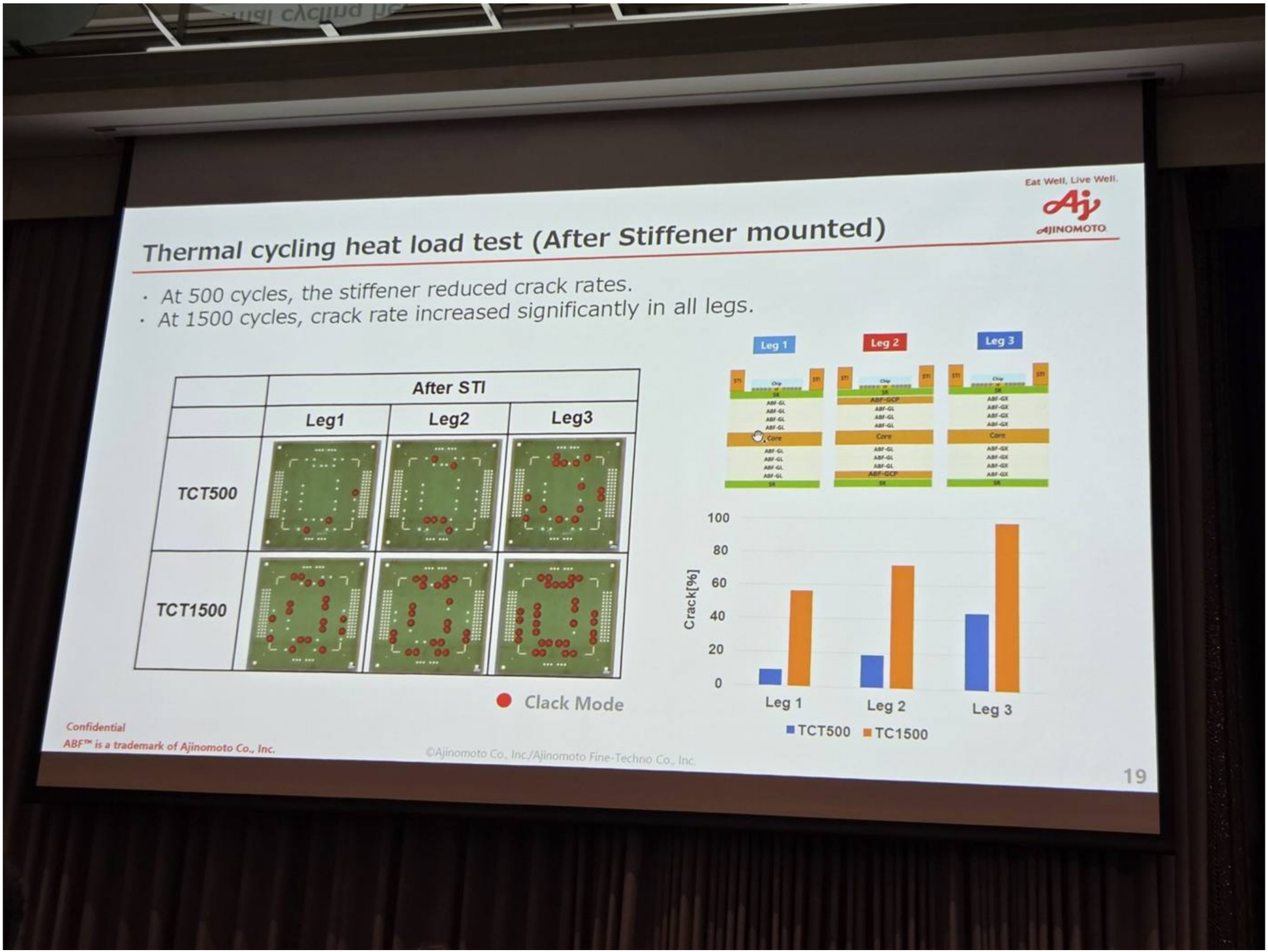
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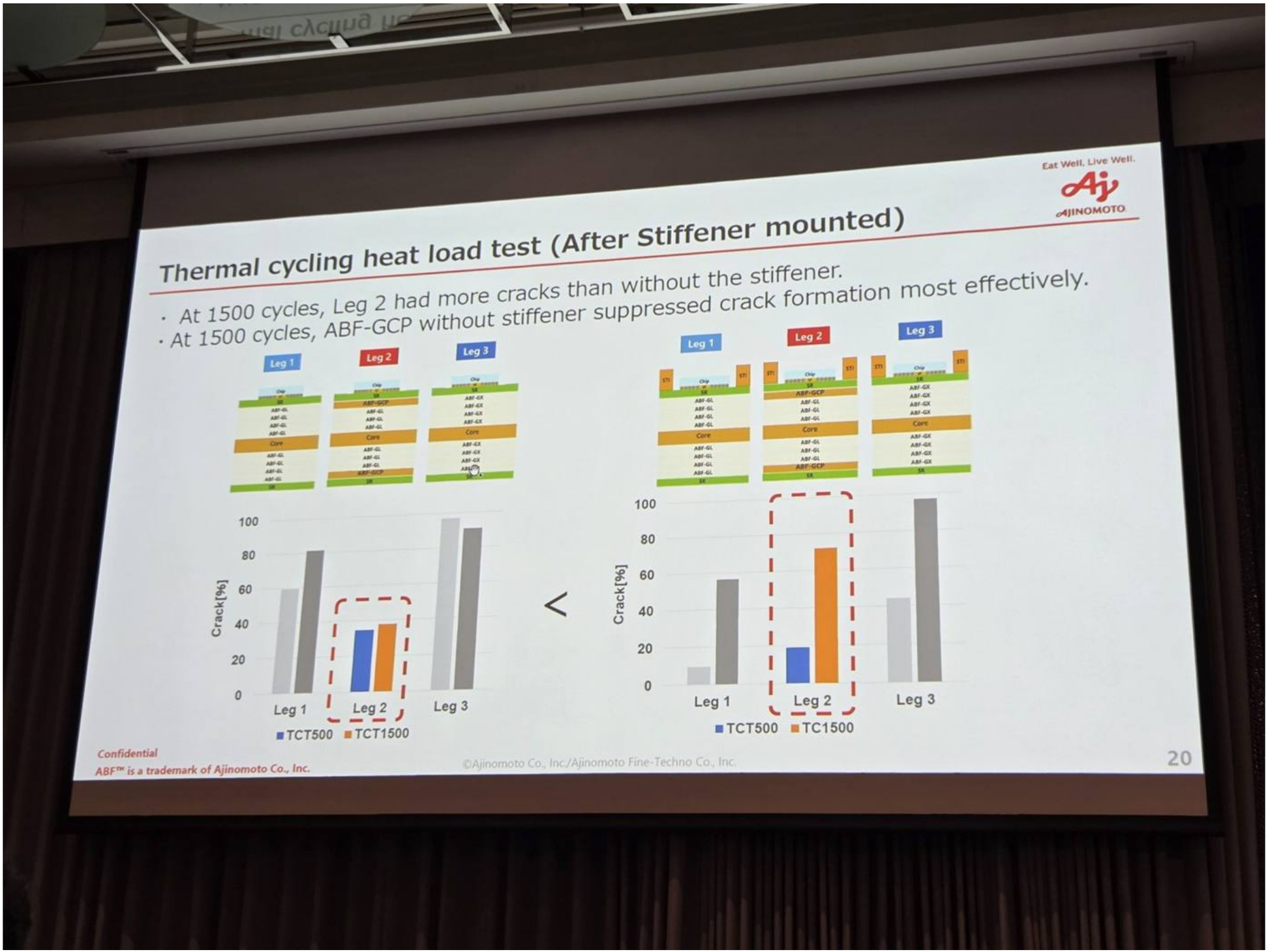
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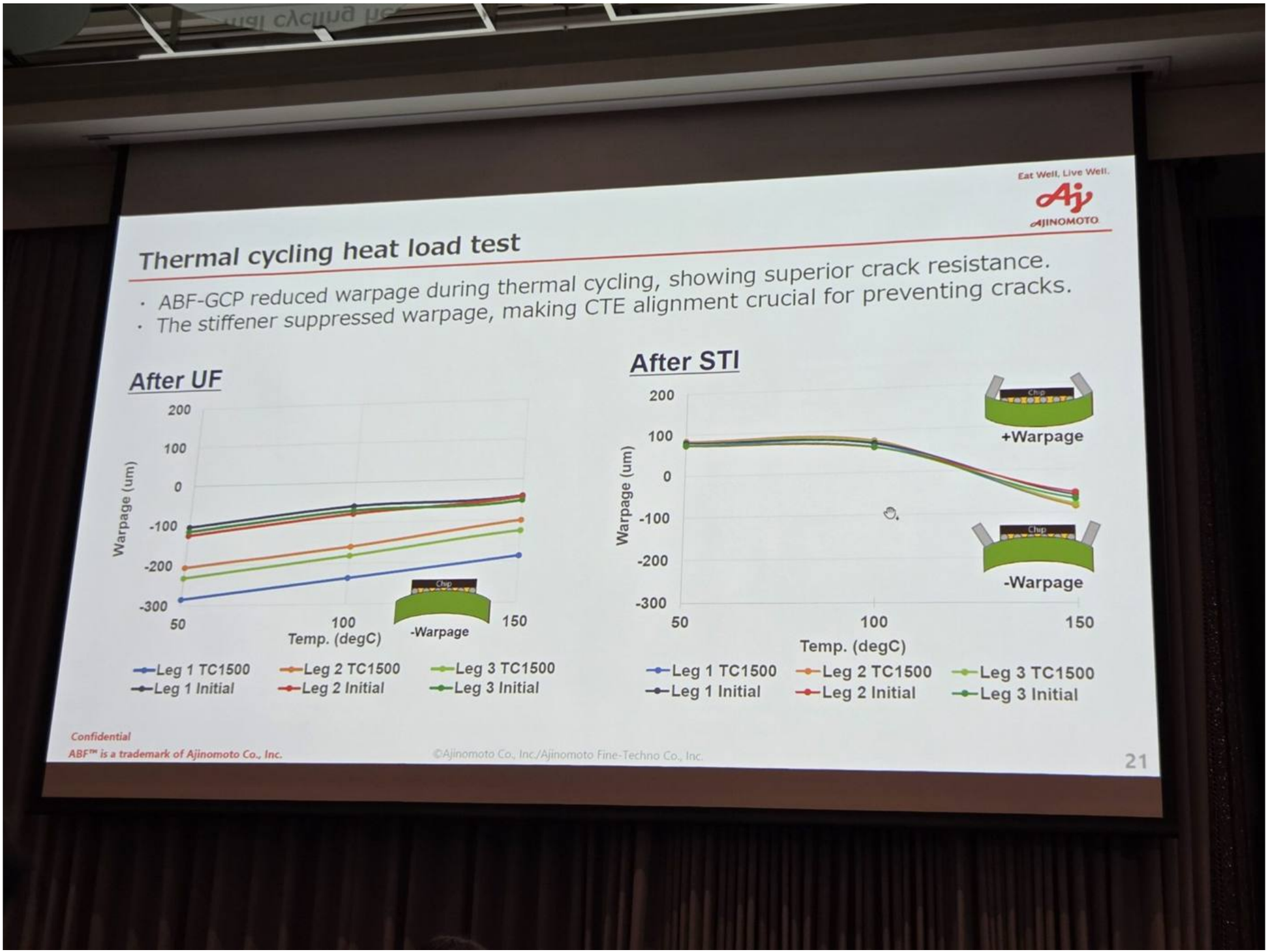
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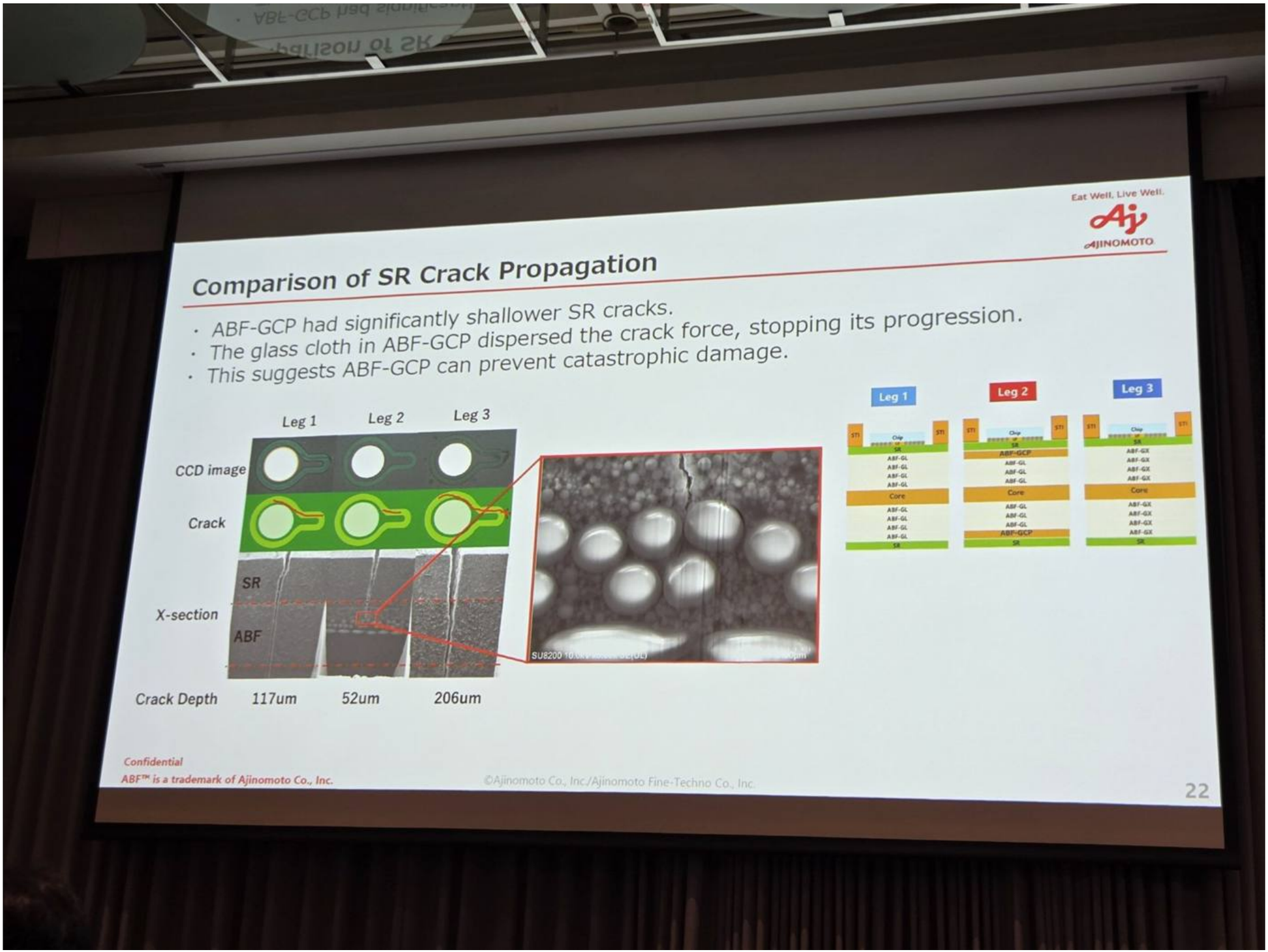
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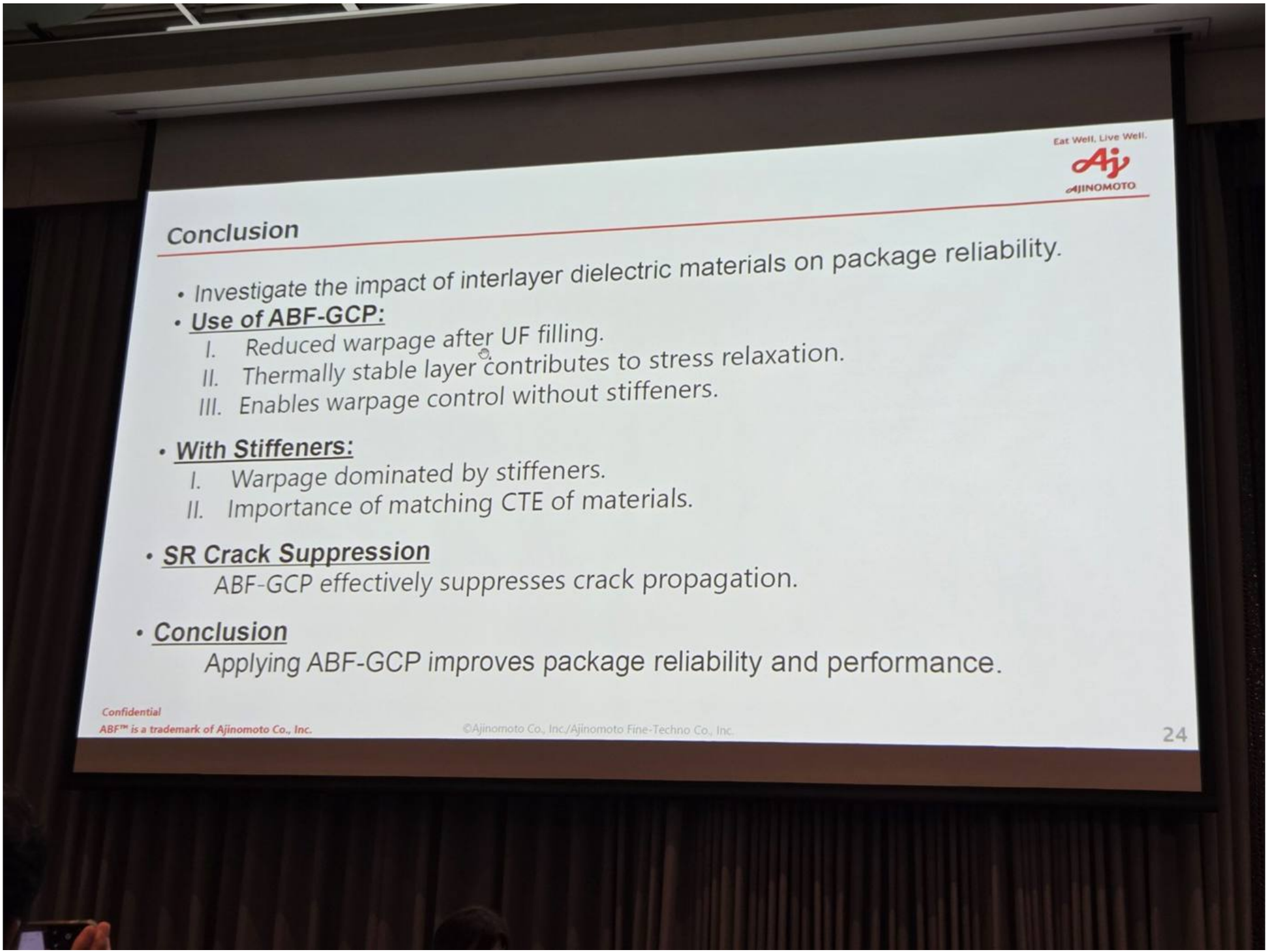
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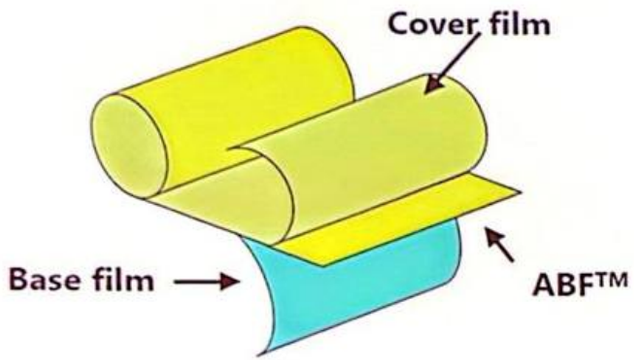
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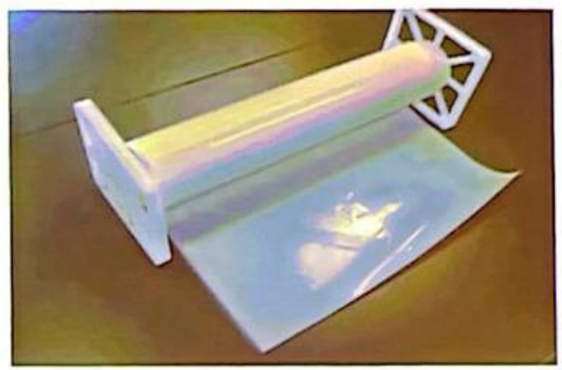



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Novel Insulation Materials for Reducing Warpage and Enhancing Reliability in Advanced Semiconductor Packaging
Minoru Sasaki
Ajinomoto Fine-Techno Co., Inc.
Nov.5, 2026
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Ajinomoto Build-up Film™ (ABF™)





➢ ABF™ Application


PC


Server



Network


AI


Automotive


Game


Mobile

(Unless otherwise stated, the word 'ABF' has the same meaning as ABF™ in this presentation file)

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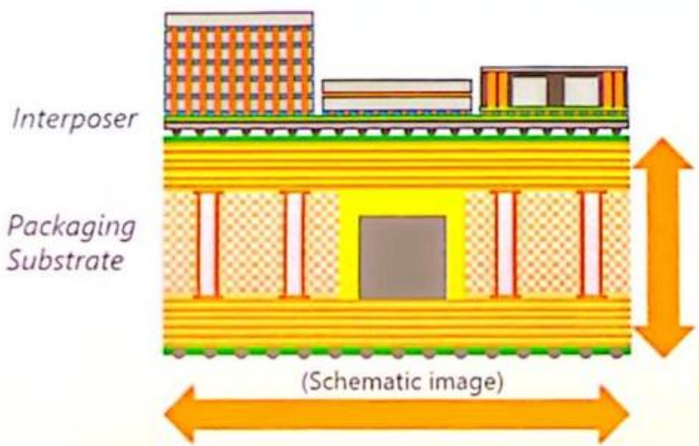
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Material Demands for Advanced Packaging

➢ Required characteristics of ABF™

Advanced Packaging Substrate



Requirement for ABF™ characteristics

Low Insertion Loss	Low Df, Low Dk
Higher Layer Count	Low CTE High Stiffness Low Stress
Larger Form Factor	
High Density Wiring	Fine Line / Fine via Capability

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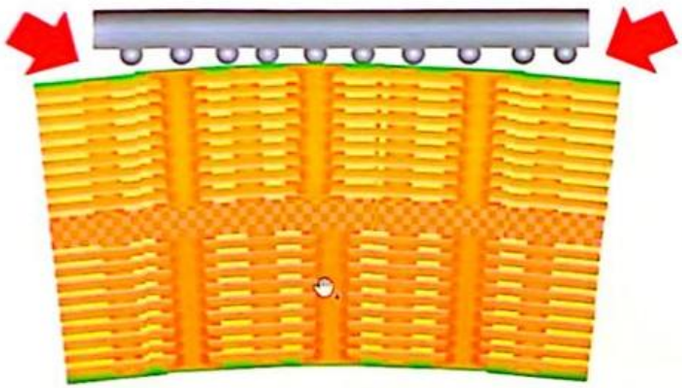
T21 유리섬유 보강 ABF의 휨과 크랙 제어

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Challenges of Large and High Layer Packaging

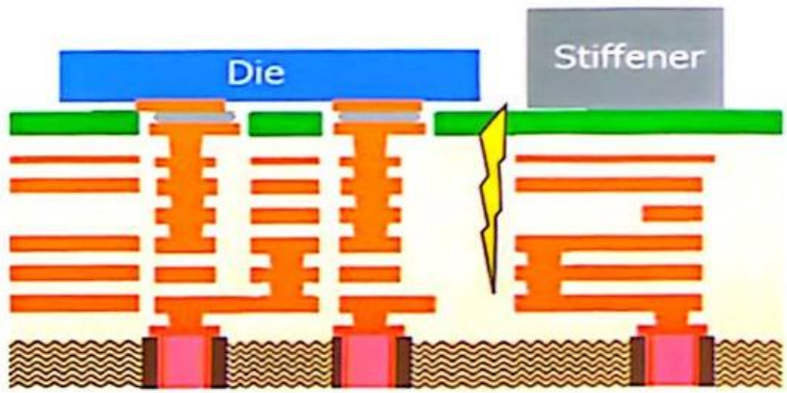
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warping



Increased Warpage Due to PCB Upscaling:
A Challenge for High-Density Integration

cracking



Crack Formation Due to Increased
Stress from Thermal Cycling

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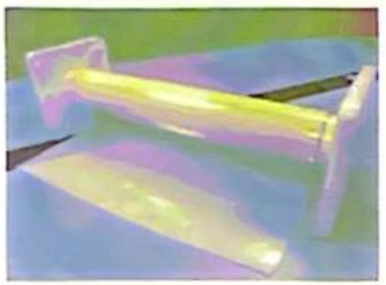
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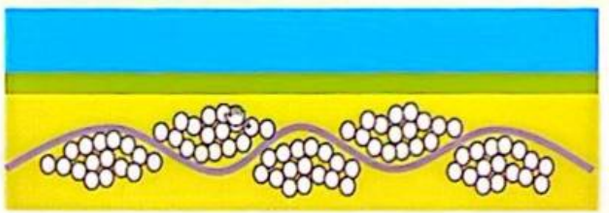
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ABF-GCP : ABF- Glass cloth with primer

※ABF-GCP : ABF- Glass cloth with primer

ABF-GCP



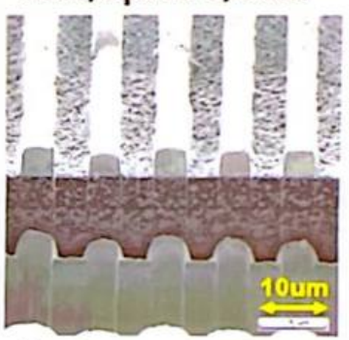


Base film
Primer
Prepreg
Cover Film

Suitability for Semi-Additive Process

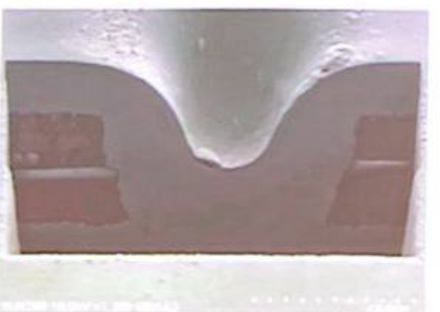
☑ Fine Line Formation
☑ Via Formation

Line/Space 5/5um



Line/Space = 5um/5um

Laser via



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T21 유리섬유 보강 ABF의 휨과 크랙 제어

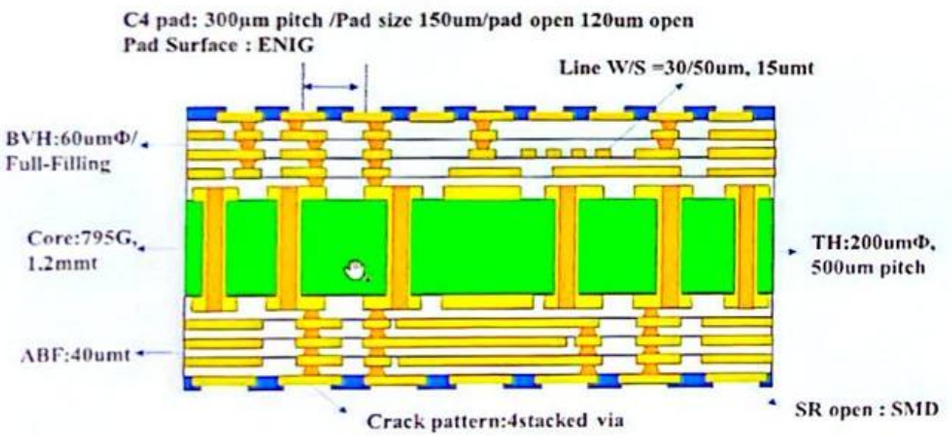
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Configuration of the substrates

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- Leg 1 : ABF GL series (Low CTE type) / All layers
- Leg 2 : ABF GCP / Outermost layer + ABF GL series / Inner layer
- Leg 3 : ABF GX series (Conventional type) / All layers

Characteristics	Unit	ABF-GL	ABF-GCP	ABF-GX
CTE x-y (25-150 degC) (Tensile TMA)	ppm/ degC	20	15	39
CTE x-y (150-240 degC) (Tensile TMA)	ppm/ degC	49	12	117





ABF GL (Low CTE)

ABF-GCP + ABF GL

ABF GX (Conventional)

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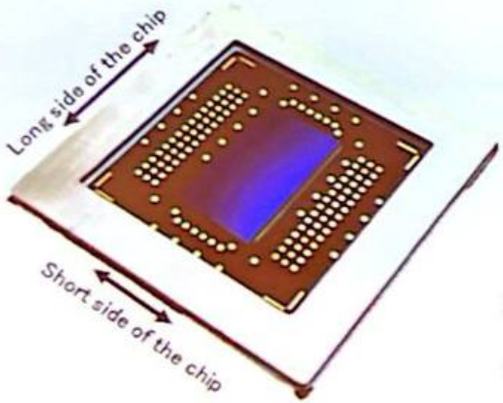
9

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Configuration of the package substrates

- Package size: 60mm x 60mm.
- Mounted a 9mm x 5mm rectangular chip to induce stress concentration.
- Attached a 2.5mm thick Cu stiffener.

Over-view image



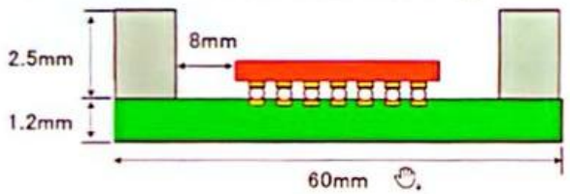
Substrate:60x60mm2
MCL-E-795G 1.2mmt

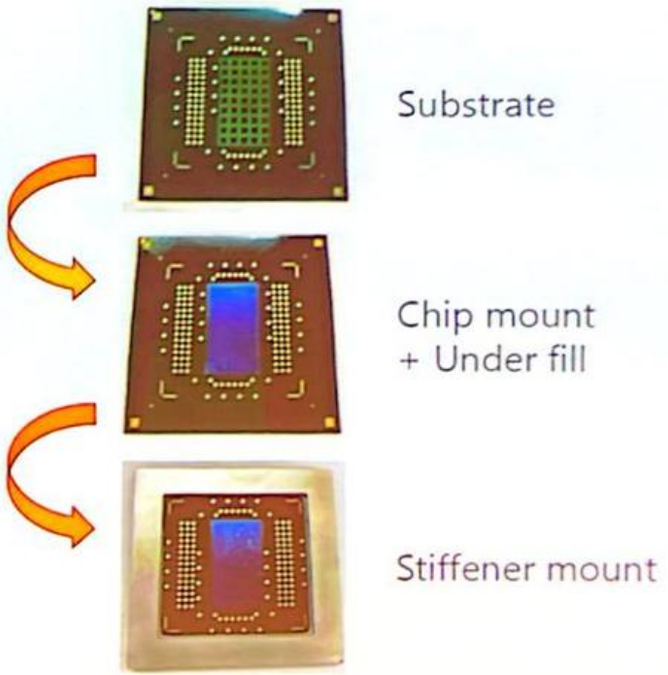
Stiffener:59x59mm2 t2.5
Material : Cu (C1000 1/2H w/CNC)
Open area:43x43mm

STAC TEG:27x15mm 550umt
Chips :9x5

Underfill UF8410-302

Cross-section view image





Substrate

Chip mount
+ Under fill

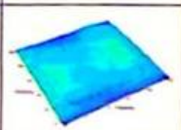
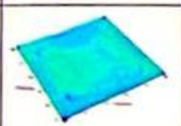
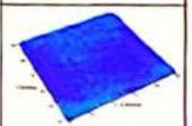
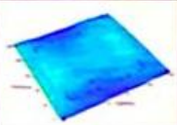
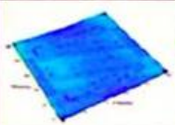
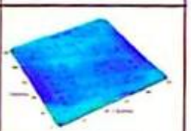
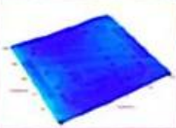
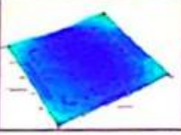
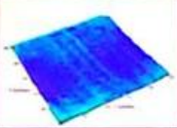
Stiffener mount

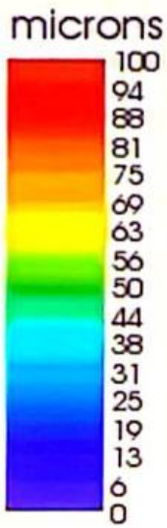
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Warpage measurement / Only substrate

- Warpage behavior was evaluated using the shadow moiré method.
- No significant warpage was observed.

	Substrate		
Leg	1	2	3
50degC			
150degC			
250degC			





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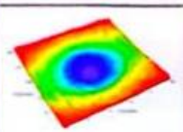
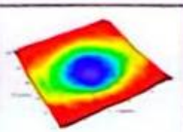
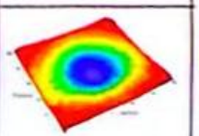
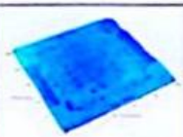
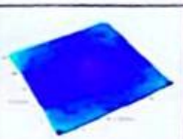
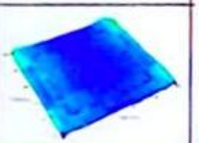
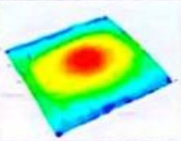
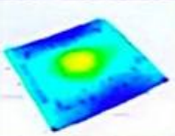
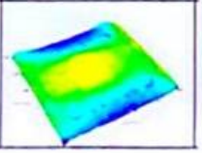
T21 유리섬유 보강 ABF의 휨과 크랙 제어

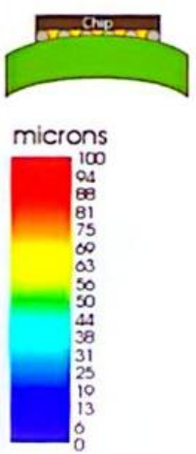
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Warpage measurement / Chip mount & Underfill

Eat Well, Live Well.

- With the chip mounted, significant warpage happened.
- Especially at 250degC, ABF-GCP could work for flatness improvement.

	After UF		
Leg	1	2	3
50degC			
150degC			
250degC			



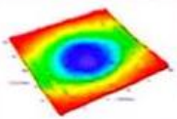
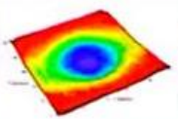
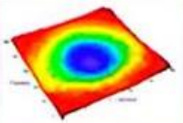
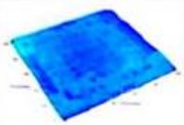
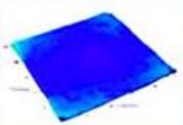
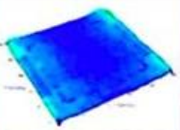
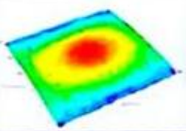
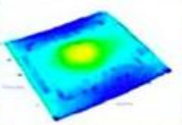
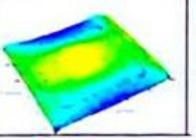


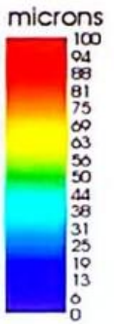
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Consideration of substrate warpage

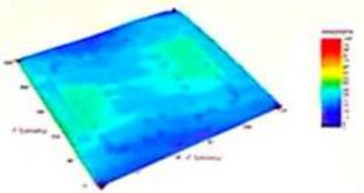
- Warpage was flattest at 150degC due to Underfill curing at 165degC.
- Cooling to room temperature caused significant warpage from CTE mismatch.

	After UF		
Leg	1	2	3
50degC			
150degC			
250degC			



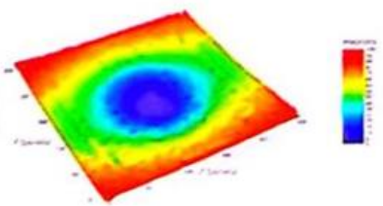
UF Curing Temp. (@165℃)





Room temp. (@25℃)





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Warpage measurement / Stiffener mount

- With the stiffener, warpage was suppressed at all temperatures.
- The stiffener's rigidity was dominant.

Leg	1	2	3
	After STI		
50degC			
150degC			
250degC			

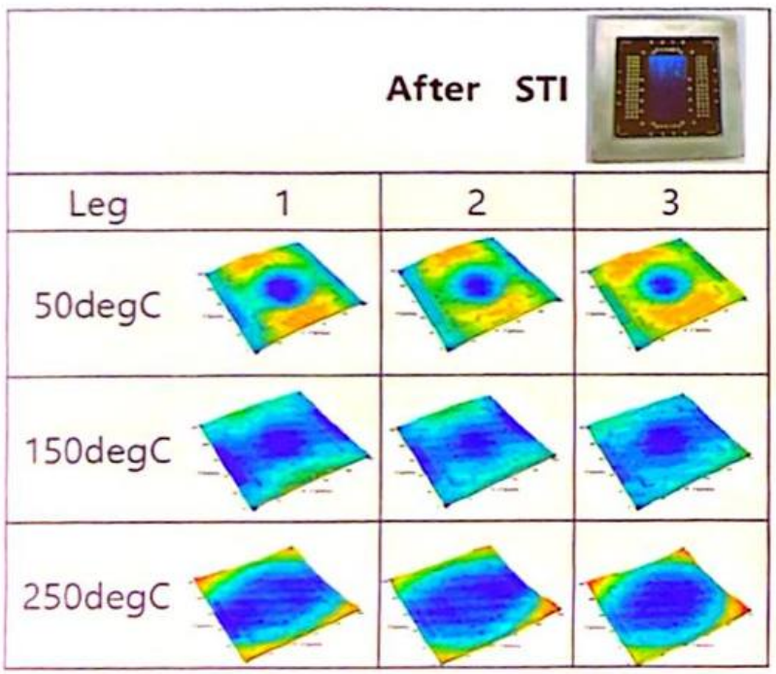


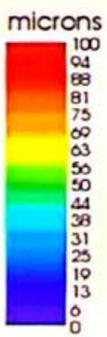
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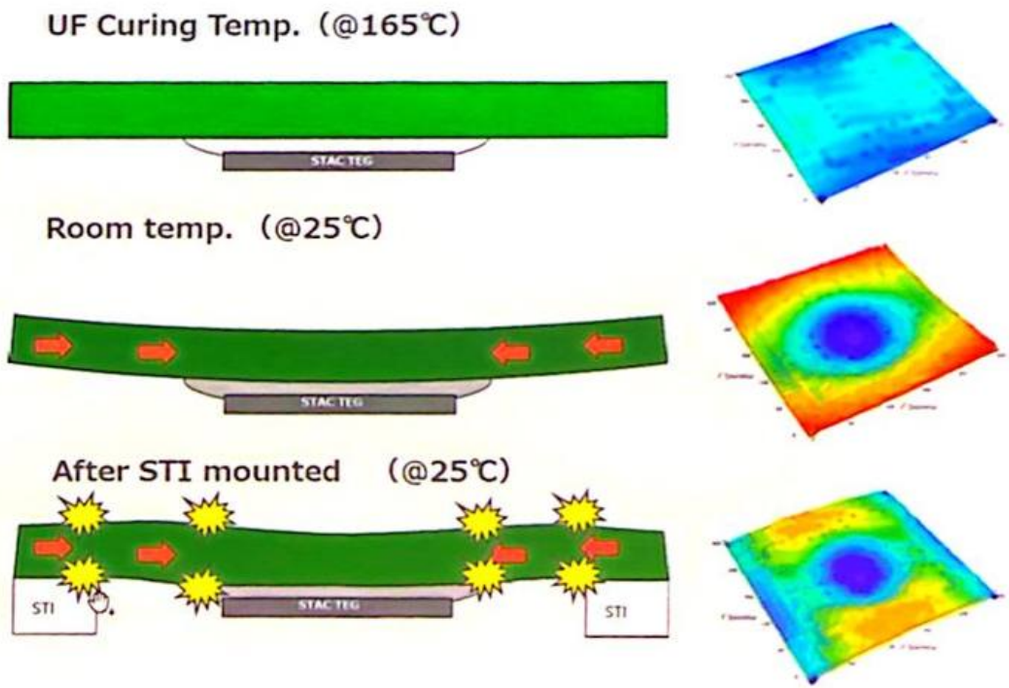
Consideration of internal stress

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- The stiffener suppresses visible warpage, but internal stress is stored due to UF-induced warpage.







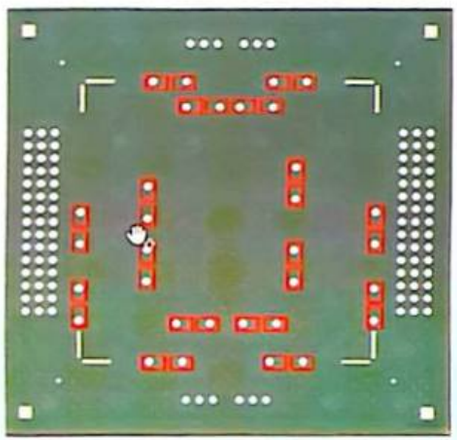
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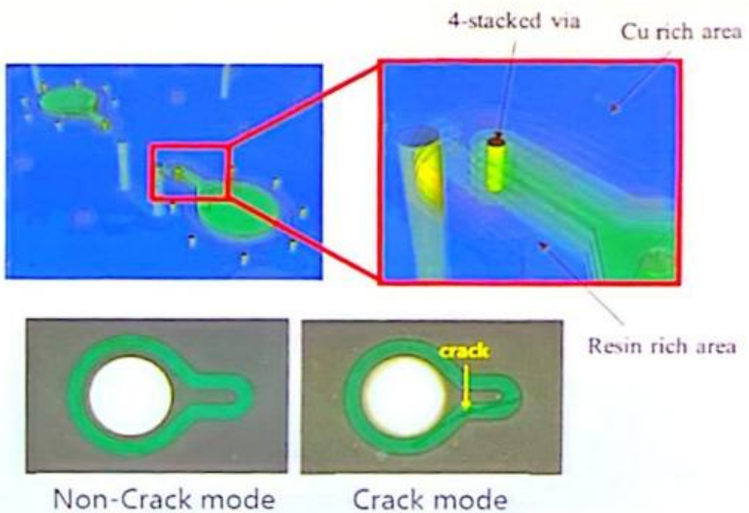
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Thermal cycling heat load test

- The substrates were subjected to thermal cycling from -65℃ to 150℃.
- Critical points prone to cracking were inspected.



32 points

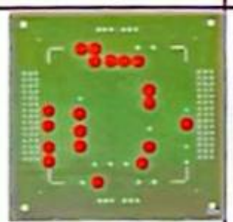
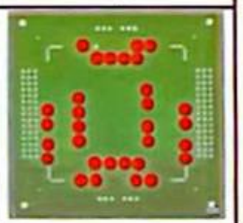
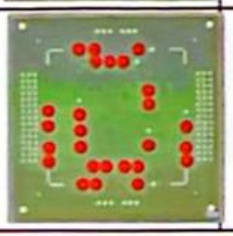
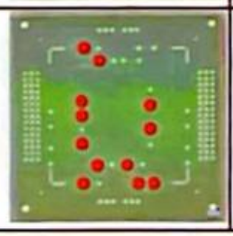
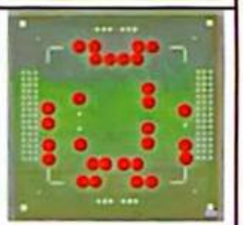


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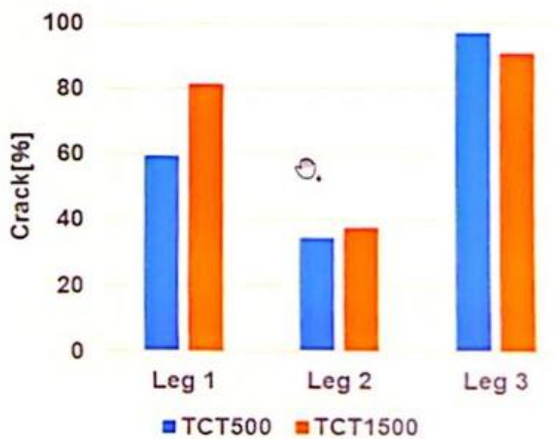
Thermal cycling heat load test (After UF)

- Results after chip mounting show Clack rates at 500 and 1500 cycles.
- ABF-GCP showed the least cracking.

	After UF		
	Leg1	Leg2	Leg3
TCT500			
TCT1500			

● Clack Mode





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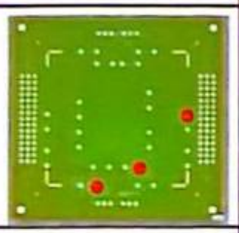
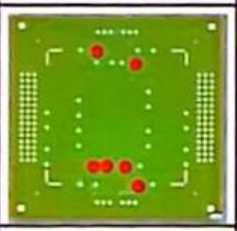
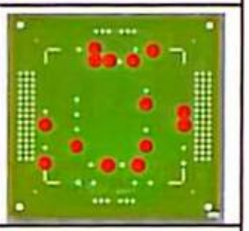
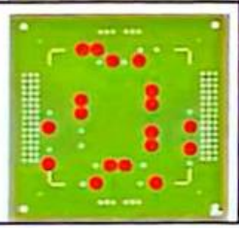
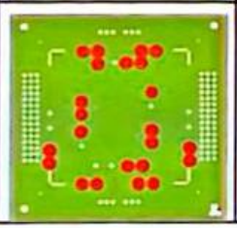
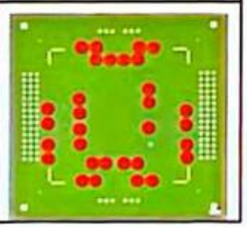
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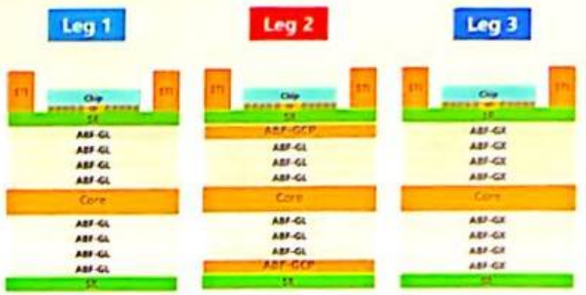
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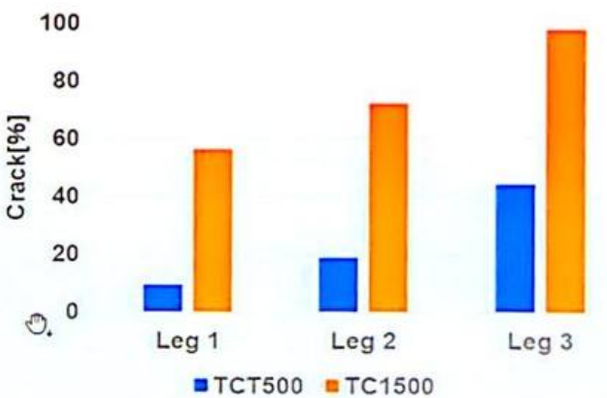
Thermal cycling heat load test (After Stiffener mounted)

- At 500 cycles, the stiffener reduced crack rates.
- At 1500 cycles, crack rate increased significantly in all legs.

	After STI		
	Leg1	Leg2	Leg3
TCT500			
TCT1500			

● Clack Mode





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T21 유리섬유 보강 ABF의 휨과 크랙 제어

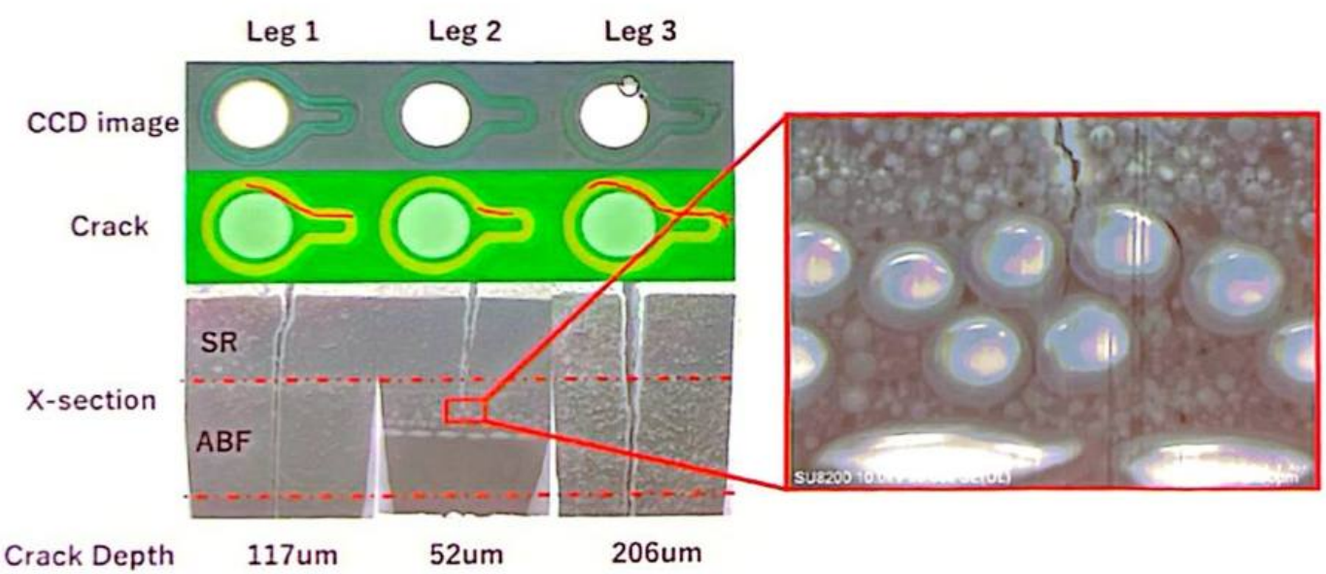
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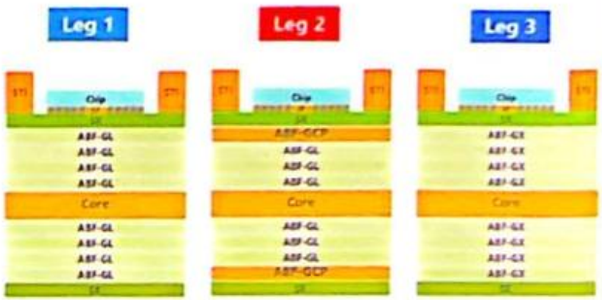
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Comparison of SR Crack Propagation

- ABF-GCP had significantly shallower SR cracks.
- The glass cloth in ABF-GCP dispersed the crack force, stopping its progression.
- This suggests ABF-GCP can prevent catastrophic damage.





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